

# Fundamentals of Photovoltaics and solar cell technology

Complete student lesson for course code **6501C**.

Revision 2026    Semester 6    Open Elective    4 modules

## Course snapshot

**Scheme:** 4 (L:4, T:0, P:0)

**Credits:** 4

**Contact:** 60 periods per semester

**Module hours listed:** 58

## Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

**Source handling:** Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

**Transparency note:** The supplied PDF does not specify a separate CIE/ESE distribution. Any limitation in the supplied PDF is not silently corrected.

## How to Study This Lesson

### Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

**Study principle:** Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

## Course Dashboard

Course code

**6501C**

Semester

**6**

Credits

**4**

Teaching scheme

**4 (L:4, T:0, P:0)**

### Module-hour and outcome mapping

| No. | Module                                    | Official hours | Relevant CO / level |
|-----|-------------------------------------------|----------------|---------------------|
| 1   | Renewable Energy Fundamentals             | 12             | CO1                 |
| 2   | Solar-Cell Fundamentals                   | 12             | CO2                 |
| 3   | Solar-Cell Characteristics and Efficiency | 16             | CO3                 |
| 4   | Solar-Cell and PV-Module Fabrication      | 18             | CO4                 |

## Prerequisites

- Semiconductor theory — Basic Electronics, code 2041, Semester 2.

## How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

## Objectives, Course Outcomes and CO-PO Information

### Official course objectives

1. The course aims to provide a basic understanding of photovoltaics and solar cell technology.

### Course Outcomes

1. Explain different renewable and non-renewable energy sources and their types.
2. Describe the fundamentals of solar cells.
3. Explain solar-cell performance and the factors affecting it.
4. Explain solar-cell and PV-module fabrication and formation.

### CO-PO mapping as supplied

The supplied CO-PO table is reproduced at outcome level from the PDF; blank cells are not treated as mapped.

Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

## Complete Syllabus Coverage

## Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

### Source-to-lesson coverage map

| Module   | Lesson topic cards | Source basis            | Official point units |
|----------|--------------------|-------------------------|----------------------|
| Module 1 | 4                  | Official Contents block | 4                    |
| Module 2 | 4                  | Official Contents block | 6                    |
| Module 3 | 4                  | Official Contents block | 8                    |
| Module 4 | 4                  | Official Contents block | 10                   |

Use this checklist to confirm that every official module topic is represented in the lesson.

### Complete official topic coverage checklist

| Module   | Topic code | Topic                                                        | Hours |
|----------|------------|--------------------------------------------------------------|-------|
| module-1 | M1.01      | Need of switching to Renewable Energy sources                | 2     |
| module-1 | M1.02      | Renewable and Non-renewable sources                          | 2     |
| module-1 | M1.03      | Main sources of Renewable                                    | 4     |
| module-1 | M1.04      | Applications, Advantages & Disadvantages of Renewable Energy | 4     |
| module-2 | M2.01      | Current conduction in semiconductor                          | 3     |
| module-2 | M2.02      | Formation of P-N junction of Semiconductor                   | 3     |
| module-2 | M2.03      | Materials for Opto-Electronic applications                   | 3     |
| module-2 | M2.04      | Concept of solar cell and its main elements                  | 3     |
| module-3 | M3.01      | Characteristics of Solar cells                               | 4     |
| module-3 | M3.02      | Design considerations of Solar cells                         | 4     |
| module-3 | M3.03      | Factors limiting the efficiency of solar cell                | 4     |
| module-3 | M3.04      | Impact of external parameters on solar cell performances     | 4     |
| module-4 | M4.01      | Classification of solar cells                                | 4     |

| Module   | Topic code | Topic                                           | Hours |
|----------|------------|-------------------------------------------------|-------|
| module-4 | M4.02      | Materials used for Photovoltaic Cells           | 4     |
| module-4 | M4.03      | Technologies for Photovoltaic Cells Fabrication | 5     |
| module-4 | M4.04      | Fabrication and formation of PV modules         | 5     |

## Learning Roadmap

**1. Renewable Energy Fundamentals**  
CO1 · 12 hours

**2. Solar-Cell Fundamentals**  
CO2 · 12 hours

**3. Solar-Cell Characteristics and Efficiency**  
CO3 · 16 hours

**4. Solar-Cell and PV-Module Fabrication**  
CO4 · 18 hours

The roadmap follows the order of the supplied syllabus.

### OFFICIAL MODULE 1

## Renewable Energy Fundamentals

Photovoltaic technology is part of a wider energy system in which resource availability, environmental impact, and conversion method determine sustainability.

**Hours: 12**

**CO1 · Bloom level: Understanding**

### Official source coverage for Module 1

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

s: Fundamentals of Sustainable Energy & Development

Introduction to RE – Need of switching to Renewable Energy sources, Difference between

Renewable & Non-renewable sources, Main sources – solar, wind, tidal, biomass, geothermal, Applications, Advantages & Disadvantages of Renewable Energy.

## **Point-by-point study guide**

## – Official syllabus point 1: s: Fundamentals of Sustainable Energy & Development

### Exact source point

**Syllabus text:** s: Fundamentals of Sustainable Energy & Development

### How to understand and study it

This point is an official syllabus requirement: “s: Fundamentals of Sustainable Energy & Development”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Fundamentals of Sustainable Energy & Development ് technical terms English-്  
്. ് definition ് principle ്; ് term-  
് function, difference, application ്. Diagram,  
sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



### Exact source point

**Syllabus text:** Introduction to RE – Need of switching to Renewable Energy sources, Difference between

## How to understand and study it

This point is an official syllabus requirement: “Introduction to RE – Need of switching to Renewable Energy sources, Difference between”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

## Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** syllabus point ഇന്ററോഡക്ഷൻ ടു റീ - നീഡ് ഓഫ് സ്വിച്ചിംഗ് ടു റീനോവബിൾ ഐംഗിൾ സോഴ്സ്, ഡിഫറൻസ് ബീറ്റ്വീൻ ടെക്നിക്കൽ ടേർംസ് എംഗ്ലിഷ്-ഇന്ററോഡക്ഷൻ ടു റീനോവബിൾ ഐംഗിൾ സോഴ്സ്. ഡിഫറൻസ് ഡിഫറൻസ് ഡിഫറൻസ്; ടെക്നിക്കൽ ടേർംസ്-ഇന്ററോഡക്ഷൻ ടു റീനോവബിൾ ഐംഗിൾ സോഴ്സ്. ഡിഫറൻസ്, ഡിഫറൻസ്, ഡിഫറൻസ്, ഡിഫറൻസ്. ഡിഫറൻസ്, ഡിഫറൻസ്, ഡിഫറൻസ്, ഡിഫറൻസ്. ഡിഫറൻസ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



## – Official syllabus point 3: Renewable & Non-renewable sources, Main sources – solar, wind, tidal, biomass

### Exact source point

**Syllabus text:** Renewable & Non-renewable sources, Main sources – solar, wind, tidal, biomass

### How to understand and study it

This point is an official syllabus requirement: “Renewable & Non-renewable sources, Main sources – solar, wind, tidal, biomass”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Renewable & Non-renewable sources, Main sources – solar, biomass ് technical terms English- ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 4: geothermal, Applications, Advantages & Disadvantages of Renewable Energy.

### Exact source point

**Syllabus text:** geothermal, Applications, Advantages & Disadvantages of Renewable Energy.

### How to understand and study it

This point is an official syllabus requirement: “geothermal, Applications, Advantages & Disadvantages of Renewable Energy.”. Prepare a comparison using the same criteria for every item: principle, performance, cost or complexity, limitation, and application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് geothermal, Applications, Advantages & Disadvantages of Renewable Energy. ് technical terms English- ്. ് definition ് principle ്; ് ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

### Learning goals

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### Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

### Concept and explanation

**Scope.** The official syllabus places **Need of switching to Renewable Energy sources** in this topic. The required scope is: Need for switching to renewable energy sources.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic engineering.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** Need of switching to Renewable Energy sources topic purpose, working principle, components variables, performance safety checks. Technical terms English-; diagram step sequence process.

### Study points

- Define Need of switching to Renewable Energy sources using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Need of switching to Renewable Energy sources is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Need of switching to Renewable Energy sources\*\***, prepare a four-column revision note with **\*purpose\***, **\*principle/sequence\***, **\*important parameters\***, and **\*application or limitation\***. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Need of switching to Renewable Energy sources without explaining the working sequence, variables, or engineering reason for each item.



## Concept and explanation

**Scope.** The official syllabus places **Renewable and Non-renewable sources** in this topic. The required scope is: Difference between renewable and non-renewable sources.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic engineering.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** Renewable and Non-renewable sources topic purpose, working principle, components variables, performance safety checks. Technical terms English-; diagram step sequence process.

## Study points

- Define Renewable and Non-renewable sources using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Renewable and Non-renewable sources is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Renewable and Non-renewable sources\*\***, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Renewable and Non-renewable sources without explaining the working sequence, variables, or engineering reason for each item.

## Concept and explanation

**Scope.** The official syllabus places **Main sources of Renewable** in this topic. The required scope is: Solar, wind, tidal, biomass and other main renewable sources.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic engineering.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** lang="ml">Main sources of Renewable topic പഠിക്കേണ്ടതാണ്. പഠിക്കേണ്ടതാണ് purpose, working principle, പഠിക്കേണ്ടതാണ് components പഠിക്കേണ്ടതാണ് variables, പഠിക്കേണ്ടതാണ് performance പഠിക്കേണ്ടതാണ് safety checks പഠിക്കേണ്ടതാണ് പഠിക്കേണ്ടതാണ്. Technical terms English-പഠിക്കേണ്ടതാണ് diagram പഠിക്കേണ്ടതാണ് step sequence പഠിക്കേണ്ടതാണ് process പഠിക്കേണ്ടതാണ് പഠിക്കേണ്ടതാണ്.

## Study points

- Define Main sources of Renewable using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Main sources of Renewable is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Main sources of Renewable\*\***, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Main sources of Renewable without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Applications, Advantages & Disadvantages of Renewable Energy** in this topic. The required scope is: Applications, advantages and disadvantages of renewable energy.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic engineering.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** Applications, Advantages & Disadvantages of Renewable Energy topic purpose, working principle, components variables, performance safety checks diagram step sequence process.

### Study points

- Define Applications, Advantages & Disadvantages of Renewable Energy using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Applications, Advantages & Disadvantages of Renewable Energy is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Applications, Advantages & Disadvantages of Renewable Energy\*\***, prepare a four-column revision note with **\*purpose\***, **\*principle/sequence\***, **\*important parameters\***, and **\*application or limitation\***. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Applications, Advantages & Disadvantages of Renewable Energy without explaining the working sequence, variables, or engineering reason for each item.

**Module summary:** Consolidate Renewable Energy Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.

## OFFICIAL MODULE 2

# Solar-Cell Fundamentals

A solar cell is a semiconductor p-n junction device that converts incident photon energy into electrical output.

**Hours: 12**

**CO2 · Bloom level: Understanding**

## Official source coverage for Module 2

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

:

Solar cell fundamentals:

Current conduction in semiconductor, Atomic structure of silicon, Energy band formation in semiconductor, P-Type and N-type material with silicon  
Formation of P-N junction of Semiconductor, Principles for Electron-Hole Pair generation by Photon absorption, Photo- electric effect, Photo-conductive effect and Photovoltaic effect.

Materials for Opto-Electronic applications.

Concept of solar cell, Main elements of silicon solar cell.

Explain the characteristics of photovoltaic cells and factors affecting

## Point-by-point study guide

## – Official syllabus point 1: Solar cell fundamentals

### Exact source point

**Syllabus text:** Solar cell fundamentals

### How to understand and study it

This point is an official syllabus requirement: “Solar cell fundamentals”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Solar cell fundamentals  
് technical terms English-് ്. ് definition ്  
principle ്; ് term-് function, difference, application  
്. Diagram, sequence, formula, unit ്  
് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



– **Official syllabus point 2: Current conduction in semiconductor, Atomic structure of silicon, Energy band formation in semiconductor, P-Type and N-type material with silicon**

**Exact source point**

**Syllabus text:** Current conduction in semiconductor, Atomic structure of silicon, Energy band formation in semiconductor, P-Type and N-type material with silicon

**How to understand and study it**

This point is an official syllabus requirement: “Current conduction in semiconductor, Atomic structure of silicon, Energy band formation in semiconductor, P-Type and N-type material with silicon”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

**Study sequence**

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Current conduction in semiconductor, Atomic structure of silicon, Energy band formation in semiconductor, P-Type ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

— **Official syllabus point 3: Formation of P-N junction of Semiconductor, Principles for Electron-Hole Pair generation by Photon absorption, Photo-electric effect, Photo-conductive effect and Photovoltaic effect.**

**Exact source point**

**Syllabus text:** Formation of P-N junction of Semiconductor, Principles for Electron-Hole Pair generation by Photon absorption, Photo- electric effect, Photo-conductive effect and Photovoltaic effect.

**How to understand and study it**

This point is an official syllabus requirement: “Formation of P-N junction of Semiconductor, Principles for Electron-Hole Pair generation by Photon absorption, Photo- electric effect, Photo-conductive effect and Photovoltaic effect.”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

**Study sequence**

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Formation of P-N junction of Semiconductor, Principles for Electron-Hole Pair generation by Photon absorption, Photo- electric effect, Photo-conductive effect ് technical terms English- ്. ് definition ് principle ്; ് ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 4: Materials for Opto-Electronic applications.

### Exact source point

**Syllabus text:** Materials for Opto-Electronic applications.

### How to understand and study it

This point is an official syllabus requirement: “Materials for Opto-Electronic applications.”. Connect each named application to the technical concept: identify the requirement, the component or method used, and the result obtained. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Materials for Opto-Electronic applications. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 5: Concept of solar cell, Main elements of silicon solar cell.**

**Exact source point**

**Syllabus text:** Concept of solar cell, Main elements of silicon solar cell.

**How to understand and study it**

This point is an official syllabus requirement: “Concept of solar cell, Main elements of silicon solar cell.”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

**Study sequence**

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Concept of solar cell, Main elements of silicon solar cell. ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 6: Explain the characteristics of photovoltaic cells and factors affecting

### Exact source point

**Syllabus text:** Explain the characteristics of photovoltaic cells and factors affecting

### How to understand and study it

This point is an official syllabus requirement: “Explain the characteristics of photovoltaic cells and factors affecting”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ഏ സിദ്ധ്യബ്ദം വിശദീകരിക്കുക. Explain the characteristics of photovoltaic cells, factors affecting ഏ ഏ technical terms English-വിശദീകരിക്കുക. definition വിശദീകരിക്കുക principle വിശദീകരിക്കുക; ഏ ഏ term-വിശദീകരിക്കുക function, difference, application വിശദീകരിക്കുക വിശദീകരിക്കുക. Diagram, sequence, formula, unit വിശദീകരിക്കുക വിശദീകരിക്കുക revision വിശദീകരിക്കുക.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

### Learning goals

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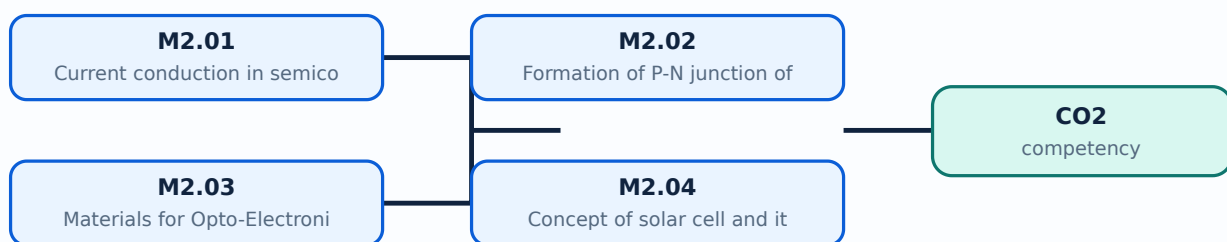
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### Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.



**Engineering / workplace application:** Current conduction in semiconductor is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Current conduction in semiconductor**, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Current conduction in semiconductor without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Formation of P-N junction of Semiconductor** in this topic. The required scope is: Formation of p-n junction in semiconductor.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** Formation of P-N junction of Semiconductor ുതം തരം purpose, working principle, ുതം തരം components ുതം തരം variables, ുതം തരം performance ുതം തരം safety checks ുതം തരം Technical terms English- ുതം തരം diagram ുതം തരം step sequence ുതം തരം process ുതം തരം.

### Study points

- Define Formation of P-N junction of Semiconductor using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Formation of P-N junction of Semiconductor is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Formation of P-N junction of Semiconductor**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Formation of P-N junction of Semiconductor without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Materials for Opto-Electronic applications** in this topic. The required scope is: Materials for optoelectronic applications.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Study frame for Materials for Opto-Electronic applications |                                                                                                                     |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Aspect                                                     | What to learn                                                                                                       |
| Purpose                                                    | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea                                               | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance                                                | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision                                       | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** Materials for Opto-Electronic applications   
 topic   
 purpose, working principle, components variables, performance   
 safety checks   
 . Technical terms   
 English-   
 diagram   
 step sequence   
 process   
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## Study points

- Define Materials for Opto-Electronic applications using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Materials for Opto-Electronic applications is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Materials for Opto-Electronic applications**, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Materials for Opto-Electronic applications without explaining the working sequence, variables, or engineering reason for each item.

## Concept and explanation

**Scope.** The official syllabus places **Concept of solar cell and its main elements** in this topic. The required scope is: Solar-cell concept and main elements.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** Malayalam support: Concept of solar cell and its main elements topic purpose, working principle, components variables, performance safety checks Technical terms English- diagram step sequence process

## Study points

- Define Concept of solar cell and its main elements using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.



**Engineering / workplace application:** Concept of solar cell and its main elements is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Concept of solar cell and its main elements**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Concept of solar cell and its main elements without explaining the working sequence, variables, or engineering reason for each item.

**Module summary:** Consolidate Solar-Cell Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.

## OFFICIAL MODULE 3

# Solar-Cell Characteristics and Efficiency

Cell performance is described by its current-voltage characteristic and parameters such as short-circuit current, open-circuit voltage, fill factor, maximum-power point, and efficiency.

**Hours: 16**

**CO3 · Bloom level: Understanding**

## Official source coverage for Module 3

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

:

Solar cell characteristics:

Current-Voltage (I-V) characteristics of a Photovoltaic cell, Power-Voltage (P-V) characteristics of a Photovoltaic cell, Equivalent circuit of a solar cell, Maximum power point (MPP).

Design considerations of Solar cells – Short circuit current ( $I_{sc}$ ), Open circuit voltage ( $V_{oc}$ ),

Fill factor (FF), Energy losses & factors for loss, Efficiency.

Factors limiting the efficiency of solar cell.

Impact of external parameters on solar cell performances – (i) Radiation,

(ii) Temperature,

(iii) Wind velocity.

Classify various photovoltaic cell technologies and describe the

## Point-by-point study guide

## – Official syllabus point 1: Solar cell characteristics

### Exact source point

**Syllabus text:** Solar cell characteristics

### How to understand and study it

This point is an official syllabus requirement: “Solar cell characteristics”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Solar cell characteristics  
് technical terms English-് ്. ് definition ്  
principle ്; ് ് term-് function, difference, application  
് ്. Diagram, sequence, formula, unit ്  
് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



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## – Official syllabus point 3: Design considerations of Solar cells – Short circuit current ( $I_{sc}$ ), Open circuit voltage ( $V_{oc}$ )

### Exact source point

**Syllabus text:** Design considerations of Solar cells – Short circuit current ( $I_{sc}$ ), Open circuit voltage ( $V_{oc}$ )

### How to understand and study it

This point is an official syllabus requirement: “Design considerations of Solar cells – Short circuit current ( $I_{sc}$ ), Open circuit voltage ( $V_{oc}$ )”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Design considerations of Solar cells – Short circuit current ( $I_{sc}$ ), Open circuit voltage ( $V_{oc}$ ) ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

– **Official syllabus point 4: Fill factor (FF), Energy losses & factors for loss, Efficiency.**

**Exact source point**

**Syllabus text:** Fill factor (FF), Energy losses & factors for loss, Efficiency.

**How to understand and study it**

This point is an official syllabus requirement: “Fill factor (FF), Energy losses & factors for loss, Efficiency.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

**Study sequence**

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Fill factor (FF), Energy losses & factors for loss, Efficiency. ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 5: Factors limiting the efficiency of solar cell.

### Exact source point

**Syllabus text:** Factors limiting the efficiency of solar cell.

### How to understand and study it

This point is an official syllabus requirement: “Factors limiting the efficiency of solar cell.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ഫാക്ടറുകൾ സോളാർ സെല്ലിന്റെ ഏകദശ്യത കുറയ്ക്കുന്നു. ് technical terms English-ഘട്ടം ഘട്ടം. ് definition ് principle ്; ് term-ഘട്ടം function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 6: Impact of external parameters on solar cell performances - (i)Radiation, (ii)Temperature

### Exact source point

**Syllabus text:** Impact of external parameters on solar cell performances - (i)Radiation, (ii)Temperature

### How to understand and study it

This point is an official syllabus requirement: “Impact of external parameters on solar cell performances - (i)Radiation, (ii)Temperature”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Impact of external parameters on solar cell performances - (i)Radiation, (ii)Temperature ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ് ്. Diagram, sequence, formula, unit ് ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



## – Official syllabus point 7: (iii) Wind velocity.

### Exact source point

**Syllabus text:** (iii) Wind velocity.

### How to understand and study it

This point is an official syllabus requirement: “(iii) Wind velocity.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് (iii) Wind velocity. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 8: Classify various photovoltaic cell technologies and describe the

### Exact source point

**Syllabus text:** Classify various photovoltaic cell technologies and describe the

### How to understand and study it

This point is an official syllabus requirement: “Classify various photovoltaic cell technologies and describe the”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Classify various photovoltaic cell technologies, describe the ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Learning goals

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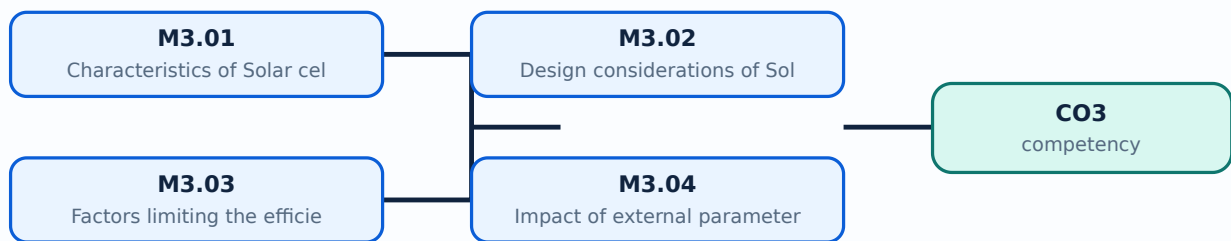
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### Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

### Concept and explanation

**Scope.** The official syllabus places **Characteristics of Solar cells** in this topic. The required scope is: Solar-cell characteristics.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Characteristics of Solar cells** എന്നു് വിഷയം പഠിക്കേണ്ടതു്. **purpose, working principle, components, variables, performance, safety checks** എന്നീ പദങ്ങൾ ഉപയോഗിച്ച് വിശദീകരിക്കുക. **Technical terms** English-ൽ എഴുതുക; **diagram** എന്നു് ചിത്രം **step sequence** എന്നു് പ്രക്രിയ പരമ്പരയെ സൂചിപ്പിക്കുന്നു.

### Study points

- Define Characteristics of Solar cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Characteristics of Solar cells is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Characteristics of Solar cells\*\***, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Characteristics of Solar cells without explaining the working sequence, variables, or engineering reason for each item.



### Concept and explanation

**Scope.** The official syllabus places **Design considerations of Solar cells** in this topic. The required scope is: Solar-cell design considerations.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** Malayalam support: Design considerations of Solar cells topic purpose, working principle, components variables, performance safety checks Technical terms English- diagram step sequence process

### Study points

- Define Design considerations of Solar cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Design considerations of Solar cells is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Design considerations of Solar cells\*\***, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Design considerations of Solar cells without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Factors limiting the efficiency of solar cell** in this topic. The required scope is: Factors limiting solar-cell efficiency.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** Factors limiting the efficiency of solar cell ൽ താല്പര്യമുള്ള വിഷയം പരിശോധിക്കുക. ഈ വിഷയം പരിശോധിക്കുമ്പോൾ, purpose, working principle, components, variables, performance, safety checks എന്നിവയെക്കുറിച്ച് പരിശോധിക്കുക. Technical terms English-ൽ പരിശോധിക്കുക; diagram വരയ്ക്കുക. step sequence process പരിശോധിക്കുക.

### Study points

- Define Factors limiting the efficiency of solar cell using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Factors limiting the efficiency of solar cell is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Factors limiting the efficiency of solar cell\*\***, prepare a four-column revision note with **\*purpose\***, **\*principle/sequence\***, **\*important parameters\***, and **\*application or limitation\***. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Factors limiting the efficiency of solar cell without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Impact of external parameters on solar cell performances** in this topic. The required scope is: Impact of external parameters on solar-cell performance.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Study frame for Impact of external parameters on solar cell performances |                                                                                                                     |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Aspect                                                                   | What to learn                                                                                                       |
| Purpose                                                                  | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea                                                             | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance                                                              | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision                                                     | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** Impact of external parameters on solar cell performances topic  
purpose, working principle, components  
variables, performance safety checks English-  
 Technical terms English-  
 diagram step  
 sequence process.

## Study points

- Define Impact of external parameters on solar cell performances using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Impact of external parameters on solar cell performances is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Impact of external parameters on solar cell performances\*\***, prepare a four-column revision note with **\*purpose\***, **\*principle/sequence\***, **\*important parameters\***, and **\*application or limitation\***. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Impact of external parameters on solar cell performances without explaining the working sequence, variables, or engineering reason for each item.

**Module summary:** Consolidate Solar-Cell Characteristics and Efficiency through definitions, working principles, engineering applications, limitations, and examination revision.

#### OFFICIAL MODULE 4

## Solar-Cell and PV-Module Fabrication

Cells are classified and interconnected into modules, panels, and arrays to obtain usable voltage, current, mechanical protection, and field reliability.

**Hours: 18**

**CO4 · Bloom level: Understanding**

### Official source coverage for Module 4

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

ts:

Technologies for Photovoltaic Cells Fabrication:

Classification of solar cell, Cell size.

Single crystalline silicon cell, Polycrystalline silicon cell.

Thin film solar cell – Amorphous Silicon, Gallium Arsenide, Cadmium Telluride, Copper

Indium Gallium Diselenide.

Dye-sensitized Solar Cell (DSSC) technology, Organic solar cell technology, Quantum Dot

Solar cell technology.

Concept of PV module, PV panel, PV array and its formation.

Silicon Group and non-Silicon Group, PV cell, PV module, PV panel and PV array fabrication.

**Point-by-point study guide**

## – Official syllabus point 1: Technologies for Photovoltaic Cells Fabrication

### Exact source point

**Syllabus text:** Technologies for Photovoltaic Cells Fabrication

### How to understand and study it

This point is an official syllabus requirement: “Technologies for Photovoltaic Cells Fabrication”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Technologies for Photovoltaic Cells Fabrication ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



## – Official syllabus point 2: Classification of solar cell, Cell size.

### Exact source point

**Syllabus text:** Classification of solar cell, Cell size.

### How to understand and study it

This point is an official syllabus requirement: “Classification of solar cell, Cell size.”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Classification of solar cell, Cell size. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 3: Single crystalline silicon cell, Polycrystalline silicon cell.

### Exact source point

**Syllabus text:** Single crystalline silicon cell, Polycrystalline silicon cell.

### How to understand and study it

This point is an official syllabus requirement: “Single crystalline silicon cell, Polycrystalline silicon cell.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Single crystalline silicon cell, Polycrystalline silicon cell. ് technical terms English-് ്. ് definition ് principle ്; ് ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 4: Thin film solar cell – Amorphous Silicon, Gallium Arsenide, Cadmium Telluride, Copper

### Exact source point

**Syllabus text:** Thin film solar cell – Amorphous Silicon, Gallium Arsenide, Cadmium Telluride, Copper

### How to understand and study it

This point is an official syllabus requirement: “Thin film solar cell – Amorphous Silicon, Gallium Arsenide, Cadmium Telluride, Copper”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Thin film solar cell – Amorphous Silicon, Gallium Arsenide, Cadmium Telluride, Copper ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## — Official syllabus point 5: Indium Gallium Diselenide.

### Exact source point

**Syllabus text:** Indium Gallium Diselenide.

### How to understand and study it

This point is an official syllabus requirement: “Indium Gallium Diselenide.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Indium Gallium Diselenide. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 6: Dye-sensitized Solar Cell (DSSC) technology, Organic solar cell technology, Quantum Dot

### Exact source point

**Syllabus text:** Dye-sensitized Solar Cell (DSSC) technology, Organic solar cell technology, Quantum Dot

### How to understand and study it

This point is an official syllabus requirement: “Dye-sensitized Solar Cell (DSSC) technology, Organic solar cell technology, Quantum Dot”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Dye-sensitized Solar Cell (DSSC) technology, Organic solar cell technology, Quantum Dot ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## — Official syllabus point 7: Solar cell technology.

### Exact source point

**Syllabus text:** Solar cell technology.

### How to understand and study it

This point is an official syllabus requirement: “Solar cell technology.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Solar cell technology. ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

**– Official syllabus point 8: Concept of PV module, PV panel, PV array and its formation.**

**Exact source point**

**Syllabus text:** Concept of PV module, PV panel, PV array and its formation.

**How to understand and study it**

This point is an official syllabus requirement: “Concept of PV module, PV panel, PV array and its formation.”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

**Study sequence**

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Concept of PV module, PV panel, PV array, its formation. ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## – Official syllabus point 9: Silicon Group and non-Silicon Group, PV cell, PV module, PV panel and PV array

### Exact source point

**Syllabus text:** Silicon Group and non-Silicon Group, PV cell, PV module, PV panel and PV array

### How to understand and study it

This point is an official syllabus requirement: “Silicon Group and non-Silicon Group, PV cell, PV module, PV panel and PV array”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് Silicon Group, non-Silicon Group, PV cell, PV module ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.



## – Official syllabus point 10: fabrication.

### Exact source point

**Syllabus text:** fabrication.

### How to understand and study it

This point is an official syllabus requirement: “fabrication.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ് syllabus point ് fabrication. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Learning goals

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### Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

### Concept and explanation

**Scope.** The official syllabus places **Classification of solar cells** in this topic. The required scope is: Classification and cell size; silicon and thin-film families.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Study frame for Classification of solar cells |                                                                                                                     |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Aspect                                        | What to learn                                                                                                       |
| Purpose                                       | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea                                  | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance                                   | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision                          | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** Classification of solar cells   
topic   
purpose, working principle,   
components   
variables,   
performance   
safety checks   
Technical terms   
English-  
; diagram   
step sequence   
process   
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## Study points

- Define Classification of solar cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Classification of solar cells is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **\*\*Classification of solar cells\*\***, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Classification of solar cells without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Materials used for Photovoltaic Cells** in this topic. The required scope is: Single-crystalline and polycrystalline silicon; amorphous silicon, GaAs, CdTe, CIGS, DSSC, organic and quantum-dot materials.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** Materials used for Photovoltaic Cells topic purpose, working principle, components variables, performance safety checks English- diagram step sequence process

### Study points

- Define Materials used for Photovoltaic Cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.



**Engineering / workplace application:** Materials used for Photovoltaic Cells is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Materials used for Photovoltaic Cells**, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Materials used for Photovoltaic Cells without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Technologies for Photovoltaic Cells Fabrication** in this topic. The required scope is: Photovoltaic-cell fabrication technologies.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Technologies for Photovoltaic Cells Fabrication** എന്ന തീർപ്പ് പരസ്യം പരസ്യം purpose, working principle, പരസ്യം components പരസ്യം variables, പരസ്യം performance പരസ്യം safety checks പരസ്യം പരസ്യം പരസ്യം. Technical terms English-ൽ പരസ്യം പരസ്യം; diagram പരസ്യം step sequence പരസ്യം process പരസ്യം പരസ്യം.

### Study points

- Define Technologies for Photovoltaic Cells Fabrication using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Technologies for Photovoltaic Cells Fabrication is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Technologies for Photovoltaic Cells Fabrication**, prepare a four-column revision note with *\*purpose\**, *\*principle/sequence\**, *\*important parameters\**, and *\*application or limitation\**. Use the official scope above as the checklist and verify that no listed term is omitted.

**Common mistake:** A common mistake is to write only a list for Technologies for Photovoltaic Cells Fabrication without explaining the working sequence, variables, or engineering reason for each item.

### Concept and explanation

**Scope.** The official syllabus places **Fabrication and formation of PV modules** in this topic. The required scope is: PV cell, module, panel and array concepts and formation; silicon and non-silicon groups.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Malayalam support:** **Malayalam support:** Fabrication and formation of PV modules   
 topic   
 purpose, working principle, components variables, performance   
 safety checks   
 . Technical terms   
 English-  
 diagram   
 step sequence   
 process   
 .

## Study points

- Define Fabrication and formation of PV modules using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

**Engineering / workplace application:** Fabrication and formation of PV modules is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

**Illustrative example:** Study example: For **Fabrication and formation of PV modules**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

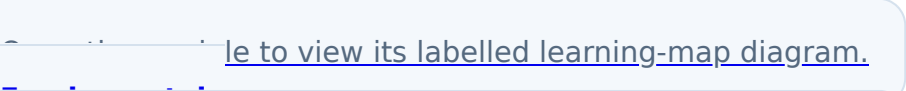
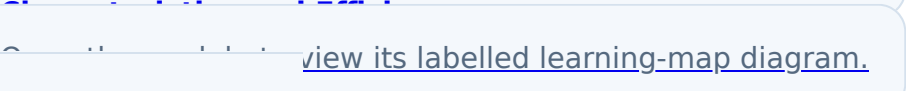
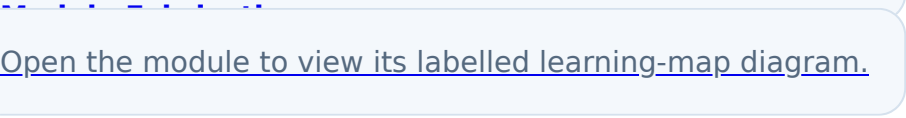
**Common mistake:** A common mistake is to write only a list for Fabrication and formation of PV modules without explaining the working sequence, variables, or engineering reason for each item.

**Module summary:** Consolidate Solar-Cell and PV-Module Fabrication through definitions, working principles, engineering applications, limitations, and examination revision.

## Formula and Notation Bank

**Formula note:** The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

## Diagram Library for Rapid Revision

|                                                                                     |                                              |
|-------------------------------------------------------------------------------------|----------------------------------------------|
|   |                                              |
|  | <a href="#">Module 2: Solar-Cell</a>         |
|  | <a href="#">Module 3: Solar-Cell</a>         |
|  | <a href="#">Module 4: Solar-Cell and PV-</a> |
|  |                                              |

## Official Activities and Practical Requirements

## Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, diagrams, and module questions in this lesson for structured study.

## Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

## Assessment Methodology

The supplied PDF does not specify a separate CIE/ESE distribution.

- The supplied PDF lists Series Test – I and Series Test – II entries, but a complete official CIE/ESE mark distribution and question-paper pattern are not specified in the supplied PDF.
- The practice model paper in this lesson is therefore explicitly a study aid, not an official examination paper.

## Module-wise Questions and Answers

### module-1 — Renewable Energy Fundamentals

#### 1. Explain Need of switching to Renewable Energy sources. [3 marks]

**Scope.** The official syllabus places *Need of switching to Renewable Energy sources* in this topic. The required scope is: Need for switching to renewable energy sources.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

<caption>Study frame for Need of switching to Renewable Energy sources</caption>

<thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in photovoltaic engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p><strong>Technical treatment.</strong> Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 2. Explain Renewable and Non-renewable sources. [3 marks]

<p><strong>Scope.</strong> The official syllabus places <em>Renewable and Non-renewable sources</em> in this topic. The required scope is: Difference between renewable and non-renewable sources.</p> <p><strong>Concept and method.</strong> Study this topic as a sequence of <strong>purpose → principle → components or variables → operation → performance or safety check</strong>. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Renewable and Non-renewable sources</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in photovoltaic engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p><strong>Technical treatment.</strong> Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

### 3. Explain Main sources of Renewable. [3 marks]

**Scope.** The official syllabus places **Main sources of Renewable** in this topic. The required scope is: Solar, wind, tidal, biomass and other main renewable sources.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic engineering.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

### 4. Explain Applications, Advantages & Disadvantages of Renewable Energy. [3 marks]

**Scope.** The official syllabus places **Applications, Advantages & Disadvantages of Renewable Energy** in this topic. The required scope is: Applications, advantages and disadvantages of renewable energy.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an



examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

</p> <div class="table-wrap"><table><caption>Study frame for Applications, Advantages & Disadvantages of Renewable Energy</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in photovoltaic engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p><strong>Technical treatment.</strong> Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## module-2 — Solar-Cell Fundamentals

### 5. Explain Current conduction in semiconductor. [3 marks]

<p><strong>Scope.</strong> The official syllabus places <em>Current conduction in semiconductor</em> in this topic. The required scope is: Current conduction in semiconductor.</p> <p><strong>Concept and method.</strong> Study this topic as a sequence of <strong>purpose → principle → components or variables → operation → performance or safety check</strong>. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Current conduction in semiconductor</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in solar-cell technology.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p><strong>Technical treatment.</strong> Relate the named subtopic to the neighbouring topics in the same

module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 6. Explain Formation of P-N junction of Semiconductor. [3 marks]

**Scope.** The official syllabus places **Formation of P-N junction of Semiconductor** in this topic. The required scope is: Formation of p-n junction in semiconductor.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Study frame for Formation of P-N junction of Semiconductor |                                                                                                                     |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Aspect                                                     | What to learn                                                                                                       |
| Purpose                                                    | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea                                               | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance                                                | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision                                       | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 7. Explain Materials for Opto-Electronic applications. [3 marks]

**Scope.** The official syllabus places **Materials for Opto-Electronic applications** in this topic. The required scope is: Materials for optoelectronic applications.

**Concept and method.** Study this topic as a

sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 8. Explain Concept of solar cell and its main elements. [3 marks]

**Scope.** The official syllabus places *Concept of solar cell and its main elements* in this topic. The required scope is: Solar-cell concept and main elements.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical**

treatment.

Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## module-3 — Solar-Cell Characteristics and Efficiency

### 9. Explain Characteristics of Solar cells. [3 marks]

**Scope.** The official syllabus places **Characteristics of Solar cells** in this topic. The required scope is: Solar-cell characteristics.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

**Study frame for Characteristics of Solar cells**

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

### 10. Explain Design considerations of Solar cells. [3 marks]

**Scope.** The official syllabus places *Design considerations of Solar cells* in this topic. The required scope is: Solar-cell design considerations.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in solar-cell technology.                         |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 11. Explain Factors limiting the efficiency of solar cell. [3 marks]

**Scope.** The official syllabus places *Factors limiting the efficiency of solar cell* in this topic. The required scope is: Factors limiting solar-cell efficiency.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect       | What to learn                                                                               |
|--------------|---------------------------------------------------------------------------------------------|
| Purpose      | Why the device, method, material, network, or process is required in solar-cell technology. |
| Working idea | How the input is converted, controlled, transported, stored, measured, or protected.        |

<th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p><strong>Technical treatment.</strong> Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 12. Explain Impact of external parameters on solar cell performances. [3 marks]

<p><strong>Scope.</strong> The official syllabus places <em>Impact of external parameters on solar cell performances</em> in this topic. The required scope is: Impact of external parameters on solar-cell performance.</p> <p><strong>Concept and method.</strong> Study this topic as a sequence of <strong>purpose → principle → components or variables → operation → performance or safety check</strong>. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Impact of external parameters on solar cell performances</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in solar-cell technology.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p><strong>Technical treatment.</strong> Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.



### 13. Explain Classification of solar cells. [3 marks]

**Scope.** The official syllabus places *Classification of solar cells* in this topic. The required scope is: Classification and cell size; silicon and thin-film families.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

### 14. Explain Materials used for Photovoltaic Cells. [3 marks]

**Scope.** The official syllabus places *Materials used for Photovoltaic Cells* in this topic. The required scope is: Single-crystalline and polycrystalline silicon; amorphous silicon, GaAs, CdTe, CIGS, DSSC, organic and quantum-dot materials.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect  | What to learn |
|---------|---------------|
| Purpose | Why the       |

device, method, material, network, or process is required in photovoltaic fabrication.

| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 15. Explain Technologies for Photovoltaic Cells Fabrication. [3 marks]

**Scope.** The official syllabus places **Technologies for Photovoltaic Cells Fabrication** in this topic. The required scope is: Photovoltaic-cell fabrication technologies.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.



**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## 16. Explain Fabrication and formation of PV modules. [3 marks]

**Scope.** The official syllabus places **Fabrication and formation of PV modules** in this topic. The required scope is: PV cell, module, panel and array concepts and formation; silicon and non-silicon groups.

**Concept and method.** Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

**Study frame for Fabrication and formation of PV modules**

| Aspect               | What to learn                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Purpose              | Why the device, method, material, network, or process is required in photovoltaic fabrication.                      |
| Working idea         | How the input is converted, controlled, transported, stored, measured, or protected.                                |
| Performance          | Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. |
| Engineering decision | How an engineer selects a configuration, operating method, component, or maintenance action.                        |

**Technical treatment.** Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## Practice Model Question Paper

### Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Need of switching to Renewable Energy sources* with one relevant application. (5 marks)
2. **2.** Define or explain *Renewable and Non-renewable sources* with one relevant application. (5 marks)
3. **3.** Define or explain *Current conduction in semiconductor* with one relevant application. (5 marks)
4. **4.** Define or explain *Formation of P-N junction of Semiconductor* with one relevant application. (5 marks)
5. **5.** Define or explain *Characteristics of Solar cells* with one relevant application. (5 marks)
6. **6.** Define or explain *Design considerations of Solar cells* with one relevant application. (5 marks)
7. **7.** Define or explain *Classification of solar cells* with one relevant application. (5 marks)
8. **8.** Define or explain *Materials used for Photovoltaic Cells* with one relevant application. (5 marks)

**Writing advice:** Use headings, standard terminology, and labelled diagrams or procedures where relevant.

## Quick Revision

### Module-wise key points

- **Renewable Energy Fundamentals:** Consolidate Renewable Energy Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.
- **Solar-Cell Fundamentals:** Consolidate Solar-Cell Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.
- **Solar-Cell Characteristics and Efficiency:** Consolidate Solar-Cell Characteristics and Efficiency through definitions, working principles, engineering applications, limitations, and examination revision.
- **Solar-Cell and PV-Module Fabrication:** Consolidate Solar-Cell and PV-Module Fabrication through definitions, working principles, engineering applications, limitations, and examination revision.

### Exam checklist

- Write the official terminology before adding explanation.

- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

**Last-minute revision:** Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

## Glossary

### Technical glossary

| Term                                                                    | Short meaning                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Need of switching to Renewable Energy sources</b>                    | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Need of switching to Renewable Energy sources&lt;/em&gt; in this topic. The required scope is: Need for switching to renewable energy sources.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as</p>    |
| <b>Renewable and Non-renewable sources</b>                              | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Renewable and Non-renewable sources&lt;/em&gt; in this topic. The required scope is: Difference between renewable and non-renewable sources.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a</p>    |
| <b>Main sources of Renewable</b>                                        | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Main sources of Renewable&lt;/em&gt; in this topic. The required scope is: Solar, wind, tidal, biomass and other main renewable sources.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a seq</p>    |
| <b>Applications, Advantages &amp; Disadvantages of Renewable Energy</b> | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Applications, Advantages &amp; Disadvantages of Renewable Energy&lt;/em&gt; in this topic. The required scope is: Applications, advantages and disadvantages of renewable energy.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and m</p>          |
| <b>Current conduction in semiconductor</b>                              | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Current conduction in semiconductor&lt;/em&gt; in this topic. The required scope is: Current conduction in semiconductor.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a sequence of &lt;stron</p> |
| <b>Formation of P-N junction of Semiconductor</b>                       | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Formation of P-N junction of Semiconductor&lt;/em&gt; in this topic. The required scope is: Formation of p-n junction in semiconductor.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a sequ</p>    |
| <b>Materials for Opto-Electronic applications</b>                       | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Materials for Opto-Electronic applications&lt;/em&gt; in this topic. The required scope is: Materials for optoelectronic applications.&lt;/p&gt;           &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a sequ</p>     |

| Term                                                            | Short meaning                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Concept of solar cell and its main elements</b>              | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Concept of solar cell and its main elements&lt;/em&gt; in this topic. The required scope is: Solar-cell concept and main elements.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a sequence</p>        |
| <b>Characteristics of Solar cells</b>                           | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Characteristics of Solar cells&lt;/em&gt; in this topic. The required scope is: Solar-cell characteristics.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a sequence of &lt;strong&gt;purpose → pr</p> |
| <b>Design considerations of Solar cells</b>                     | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Design considerations of Solar cells&lt;/em&gt; in this topic. The required scope is: Solar-cell design considerations.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a sequence of &lt;strong&gt;</p> |
| <b>Factors limiting the efficiency of solar cell</b>            | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Factors limiting the efficiency of solar cell&lt;/em&gt; in this topic. The required scope is: Factors limiting solar-cell efficiency.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a seque</p>       |
| <b>Impact of external parameters on solar cell performances</b> | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Impact of external parameters on solar cell performances&lt;/em&gt; in this topic. The required scope is: Impact of external parameters on solar-cell performance.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt;</p>       |
| <b>Classification of solar cells</b>                            | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Classification of solar cells&lt;/em&gt; in this topic. The required scope is: Classification and cell size; silicon and thin-film families.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a</p>       |
| <b>Materials used for Photovoltaic Cells</b>                    | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Materials used for Photovoltaic Cells&lt;/em&gt; in this topic. The required scope is: Single-crystalline and polycrystalline silicon; amorphous silicon, GaAs, CdTe, CIGS, DSSC, organic and quantum-dot mate</p>                               |
| <b>Technologies for Photovoltaic Cells Fabrication</b>          | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Technologies for Photovoltaic Cells Fabrication&lt;/em&gt; in this topic. The required scope is: Photovoltaic-cell fabrication technologies.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and method.&lt;/strong&gt; Study this topic as a</p>       |
| <b>Fabrication and formation of PV modules</b>                  | <p>&lt;p&gt;&lt;strong&gt;Scope.&lt;/strong&gt; The official syllabus places &lt;em&gt;Fabrication and formation of PV modules&lt;/em&gt; in this topic. The required scope is: PV cell, module, panel and array concepts and formation; silicon and non-silicon groups.&lt;/p&gt; &lt;p&gt;&lt;strong&gt;Concept and m</p>             |

## Official References and Resources

### References listed in the supplied syllabus

1. N.S. Rathore, Khobragade Chetan and Asnani Bhawana, Fundamentals of Renewable Energy Sources.
2. Mehmet Kanoğlu, Yunus A. Çengel and John M. Cimbala, Fundamentals and Applications of Renewable Energy.
3. G.N. Tiwari, Fundamentals of Renewable Energy Sources.
4. Aldo Vieira da Rosa, Fundamentals of Renewable Energy Processes.

### **Official learning resources listed**

- <https://archive.nptel.ac.in/courses/115/107/115107116/>

## **In-page Search**

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